

Letter of Recommendation

10/31/2025

To Whom It May Concern

Subject: Strong Recommendation for Michael Altamirano

I am writing to offer my strongest recommendation for **Michael Altamirano**. I have known Michael for **11 weeks** as their instructor in the intensive "**Artificial Intelligence (AI) With Data Science**" program.

The program is a rigorous, 110-hour course covering everything from Python programming and data science fundamentals to advanced machine learning techniques in Python and SQL. Based on their exceptional performance, I can attest to Michael's technical proficiency, problem-solving abilities, and dedication to the data science discipline.

Mastery of Core Data Science and Programming Skills

Michael demonstrated a superior ability to quickly grasp complex concepts, particularly in the foundational data science segment of the course.

- They mastered the essentials of the Python environment, including manipulating CSV files and using the NumPy library.
- They are highly proficient in using the Pandas library for data manipulation and cleaning, including using functions like group by, merge, and creating pivot tables.
- The student is competent in running inferential statistical analyses, including explaining concepts like distributions, sampling, and T-tests.

Application of Machine Learning & AI Principles

Michael's project work showcased their ability to apply theoretical knowledge to real-world problems. They excelled in both traditional machine learning and deep learning:

1. Traditional ML and Data Analysis (Wine Quality Prediction)

For the Wine Quality Prediction project, Michael demonstrated a complete understanding of the machine learning pipeline:

- They executed comprehensive **Exploratory Data Analysis (EDA)**, including visualization of distributions, boxplots, and correlation matrices.
- They successfully handled **pre-processing** tasks, including checking for missing values,

handling outliers, and feature standardization.

- They adeptly framed the problem as both a **regression** (predicting numeric quality) and a **classification** task (labeling wine as good/bad).
- They trained multiple baseline and advanced models, including **Random Forest and Gradient Boosting** for both tasks, and evaluated them using appropriate metrics like **RMSE, R^2 , Accuracy, F1-Score, and ROC AUC**.

2. AI and Deep Learning (Flower Classification)

In the image classification project, which involved AI/Deep Learning with Python, Michael successfully built an AI-based **Convolutional Neural Network (CNN)** model using **TensorFlow and Keras**.

- They worked with the fundamental mathematical building blocks of neural networks (tensors, vectors, matrices) and completed the full CNN workflow from image preprocessing to model training.
- This project confirmed their ability to handle image data, build complex deep learning architectures, and evaluate the resulting models.

Conclusion

Michael consistently performed at the top of the class, showing the **discipline, technical skill, and commitment** necessary to excel in a professional data science or AI role. They are not merely learners of technology; they are **implementers** who can take a problem, process the data, and deliver a well-evaluated, working model.

I recommend Michael with **zero hesitation**. They possess the practical experience and theoretical foundation required to contribute immediately to your team. Please do not hesitate to contact me if you require any further information.

Sincerely,

A handwritten signature in black ink, appearing to be 'Muhammad Ahsan', with a stylized, flowing script.

Muhammad Ahsan

Course Instructor

AI and Data Science Program

AAA Institute